SECP2523 – DATABASE

SEM 1 - 2026/2027

INITIAL SYSTEM PROPOSAL

AUTISM EARLY SCREENING SUPPORT SYSTEM

Group Name: TechNova

Members:

1. **Nasaaie bin Noriskamar - A25CS0118 (Team Leader)**
2. **Balqis Batrisya binti Jalaluddin - A25CS0196**
3. **Qistina Batrisyia binti Noor Mohd Azlan - A25CS0340**
4. **Yong See En - A25CS0168**
5. **Han Tian Chou - A25CS0064**

Stakeholder: EXCELLENT SYNERGY RESOURCES

Ts. Nurasyikin Moktar	**15 April 2026**

PROPOSED SYSTEM BACKGROUND

1. Introduction

The proposed project is the Autism Early Screening Support System, a web-based database application that digitalises the early screening process as a method to ease the burden on families of young children at risk of Autism Spectrum Disorder (ASD). The process covers screening, identification, behaviour monitoring, and continuous follow-up. The system is proposed for Excellent Synergy Resources, a non-governmental organisation (NGO) with a vision of reaching out to families of autistic children and helping refer them to professional assessment.

Currently, screening results, disability registration details and follow-up notes are kept manually through paper and in separate, unconnected places. The purpose of this project is to replace scattered records with one secure and centralised database. The database will store user accounts, child profiles based on Part A of the BPPOKU Pindaan 2019 form, M-CHAT-R screening responses, behaviour logs, appointments and referral details. Through this effort, parents or caregivers can easily enter and view their child's information, and NGO staff can efficiently retrieve records and generate reports.

2. Background Study

The organisation, Excellent Synergy Resources is an NGO that supports families of children with ASD. The targeted users of the system are mainly parents or caregivers, NGO staff, and the specialists and centres that families are referred to.

Based on collected information, early screening in Malaysia usually begins at a government health clinic, where toddlers are screened with the M-CHAT-R questionnaire during routine child visits at 12-24 months. Children flagged as at risk are referred to a paediatrician, and families who wish to register for disability (OKU) must fill in the BPPOKU form by hand and submit to Jabatan Kebajikan Masyarakat counter, or apply online through MyDaftar OKU. Appointments with specialists and centres are made by phone call and a manual diary check, which takes about 1 to 3 working days.

The data management of M-CHAT-R forms and BPPOKU registration forms are conducted paper-based, and the NGO keeps hardcopy of family records. Screening results and registration details are handled separately as no centralized database exists and processes are done manually. Follow-ups depend on the staff remembering to call or message each parent, and parents rely on their own memory of the child's behaviour

incidents, often resulting in incomplete or missing details left out by the time they meet up with a doctor.

3. Problem Statement

Through conducted business process analysis and interview findings from previous phases, the following core problems were identified in the existing manual and paper-based system.

- **Disjointed and Inaccessible Process**
  Current methods of screening a child's development require excessive resources and are hard to access. As the process is fragmented, parents may find it difficult to find out what needs to be done when suspecting that their child is underdeveloped.
- **Overreliance on the Manual Screening Practices**
  Despite the availability of more advanced technologies, screening procedures remain heavily dependent on manual labor. Together with lack of communication and coordination between parties, screening often becomes delayed.
- **Existence of Systemic Obstacles and Absence of Integration**
  In most cases, there is a low degree of awareness as well as the absence of standardization. In addition, the lack of integration between the health care and education sectors leads to information being exchanged poorly between relevant parties.
- **No Technology-Based Centralization Solution**
  With no technology-based solution, data management and coordination of activities by various NGOs and parents are inefficient. Tracking results for screening tests, such as M-CHAT-R, and managing profiles based on BPPOKU guidelines are difficult to be done manually, and records can be lost or hard to retrieve.
- **Absence of Post-Therapy Support**
  After the limited monthly clinical session ends, parents are left without any structured way to track behaviour or flag concerning patterns. This concerns the post-therapy support that needs to be given towards the autistic children.

4. Proposed Solutions

The recommended solution is known as the Autism Early Screening Support System, which is an advanced Web technology system utilizing SQL as a tool for manage system database . The system combines standard medical screening procedures with daily management systems to create an efficient technological platform for the use of parents and NGOs.

4.1 Core System Infrastructure and Security

The foundation of the system lies in data security and privacy.

4.1.1 Authentication and Access Controls

The system uses safe methods for logging into and out of the system, which ensures that confidential data is protected by not allowing access to people who are not authorized to see the data.

4.1.2 Registration and Account Management

A separate module of the system is used to manage users' accounts. This module allows new users to register for an account by providing the required information such as their name, email address and password. After successful registration, users can log in to the system using their registered email address and password. The system verifies their login credentials to ensure that only authorised users can access their accounts. Users can also view and update their account information when necessary. In addition, the system provides secure account management by protecting users' personal information and password. If a user forgets their password, the system may provide a password recovery function to help them recover the access to their account.

4.1.3 Data Security and Privacy

Considering the confidentiality of data in questions, data security is a critical feature of the core infrastructure of the proposed solution, as it will serve as a reliable environment for the formative years intervention program.

4.2 Standardized Screening & Profile Management

This module digitalizes the old screening processes manually performed to guarantee accuracy and fast intervention.

4.2.1 Digital Profile Management (BPPOKU Pindaan 2019)

The program will digitalize Bahagian A (Biodata) from the BPPOKU (Pindaan 2019) form. This process will include the following important pieces of information:

- ➢ Personal Details

  Name, MyKad / MyKid identification, and contact details.

- ➢ Education & Professional Experience

  Education level and parents/guardians' employment.

- ➢ Details of Next of Kin

  The system collects detailed information about relatives close to the person for further communication in case of necessity.

4.2.2 M-CHAT-R Screening Integration

This feature includes a test with 20 questions performed according to the M-CHAT-R standard. The users provide "Yes" or "No" answers regarding children's behavior, such as do they react to a gesture or respond when called?

4.2.3 Automated Scoring Algorithm of Risk Level

The system will be able to provide risk levels automatically, making sure that each child undergoes an evaluation process that is not only consistent but also standardized and objective in nature. For those children with results falling within the Low Risk category, which would include any score of 0-2, there would not be a need to pursue any additional actions unless there are any future worries. Those with scores ranging from 3-7 are considered to have Moderate Risk, automatically prompting notification about the requirement of taking some actions to further examine their behavior. In instances where the scores range from 8-20, the result will be classified as a High Risk automatically, skipping other steps to indicate the urgent requirement of referral of the client for professional diagnostic testing.

4.3 Support, Analytics and Specialized Management

In addition to the diagnosis, this system features modules which enable monitoring and assistance on a continuous basis.

4.3.1 Automated Notification System

To resolve the issue of "delayed referrals" in the conventional approach, this module helps generate automated alerts regarding upcoming screenings or follow-up consultations.

4.3.2 Reporting and Analytics

Data analytics provide an opportunity to understand how the development of a child evolves throughout time, as visualized by parents or NGOs.

4.3.3 Behavior Monitoring

This module monitors the emotional state and mood of the child, enabling caregivers to identify any possible triggers for certain behaviors. These are some of the functions of this module:

- Behavior Log Processor

  A real-time function that enables the caregiver to log emotional status, particular behavioral instances, and the corresponding environmental antecedents.

- Trigger Analyzer

  A pattern-matching algorithm that scans historical logs to identify correlations between environmental factors that repeat themselves and behavioral disruptions that occur again.

- Mood Trend Evaluator

  A computation function that calculates and charts baseline emotional movements over specified timelines, providing parents and NGOs with empirical evidence of structural improvement.

4.3.4 Appointment Scheduling and Referral Management

This module allows parents to book appointments directly with specialist and referral centres, as well as with education centres, through the system. When a child's behavioural log or screening result shows a concerning trend, parents can make an appointment without needing to rely on a phone call, resolving the ambiguity and delay parents currently face once the limited monthly clinical sessions come to an end. The module checks real-time availability of the specialist or

referral centre, records the appointment slot, and automatically issues a referral letter and appointment reminder to the relevant party. This directly replaces the current appointment method and the phone call and manual diary check process, which currently takes 1 to 3 working days and produces no automatic reminder.

4.3.5 Educational Placement and Special Education Management

Through the 4.3.4 module, parents are allowed to search, filter, and view the physical locations of specialized education and early intervention centres within their vicinity. When a child's screening result or behavioral logs indicate a need for early learning support, parents can quickly locate nearby facilities, view available services, and access contact information without any manual search delays. This bridges the gap between clinical diagnosis and educational placement by providing a centralized directory of specialized schools.

4.3.6 Community Groups and Peer Support

An interactive communication hub provides a dedicated discussion platform where parents and caregivers can connect, share experiences, and offer mutual peer support. Users can join localized or topic-specific forums to exchange and navigate educational systems. By centralizing communication within a secure environment, system users can reduce the isolation felt by caregivers and eliminate the scattered nature of relying on external social media channels.

4.4 Feasibility Study

4.4.1 Technical Feasibility

The technical feasibility of the Autism Early Screening Support System is possible considering the team's strengths in data engineering and structured programming. The front end can be developed with HTML5 to maintain a clean format that will allow the M-CHAT-R assessment procedure to proceed efficiently. For the back end, C++ will be used together with OOP methods to provide modularity in processing data using encapsulated classes for user accounts and test results. Data quality will be ensured using SQL Relational Database, especially in protecting confidential BPPOKU (Pindaan 2019) biodata and monitoring the individual's developmental history. Project management will be implemented systematically using the Github platform for versioning and Kanban charts for visualizing tasks.

4.4.2 Operational Feasibility

Operatively, the aim is to address the existing problem of fragmentation prevalent in the manual autism screening method. By adopting the BPPOKU standard (Pindaan 2019) into a technological medium, the proposed system will conform to existing documentation practices in Malaysia's welfare and healthcare institutions, thus facilitating easier implementation by NGOs. The system is tailor-made for the unique needs of three key players: helping parents get access to a screening tool, providing children with assistance through Routine and Behavior Management, and offering NGOs an online platform for analytics. In addition to doing away with the delays associated with manual paper work, automation guarantees that the screening process becomes a dynamic and ongoing exercise for children.

4.4.3 Economical Feasibility

Economic viability of the Autism Early Screening Support System is determined through quantitative assessment of costs involved in its developmental stage against potential savings in operations. Through computerization of the M-CHAT-R test results and BPPOKU data handling process, the proposed system will create great value through cost reduction and minimizing human error. Based on the calculation below, the project is regarded as an extremely viable venture due to the high Profitability Index (PI) of 1.96, which is very much higher than the threshold value of 1.0. It implies that for every dollar invested in the system, there will be a two-fold return.

5. Objectives

The objective of the Autism Early Screening Support System is to develop a centralized database application that supports early autism screening, child profile management, and behavioural monitoring.Focussing on reducing manual paperwork, organizing records,reducing scattered child information, and providing an easier access to relevant information.

1. Develop a centralized database:

Stores and manages children's personal information, screening records, behavioural logs, and appointment details more organized in order to reduce data fragmentation.

2. Digitize the process of child profile management:

Based on the BPPOKU Amendment 2019 (Section A) requirements, allows parents/caregivers to record, view, and update children's personal and family information using the system.

3. Implement the M-CHAT-R screening module:

An automated scoring mechanism is provided and can be able to categorize the results into Low, Moderate, and High risk levels to detect potential autism risks.

4. Provide behavioural monitoring and record management:

Enable parents/caregivers to record children's emotional conditions and behavioural patterns

5. Facilitate appointment and referral management:

Keep all the appointment records, and follow-up schedules to avoid delays.

6. Generate an analytical information reports:

By referring to the stored screening records and behavioural data,it can be used to generate the data and make an analyze.The result of the analyzation process can be converted into a report.

6. Scopes

The system scope consist of:

6.1 User Management

The system allows parents/caregivers, and NGO administrators to register and log in using their accounts.Users also can manage their own account information and update their information.The system provides authentication and access control to ensure that only authorized users have an access.

6.2 Child Profile Management

Parents/caregivers can register children's profiles by fill in personal information such as name, identification number, date of birth, and contacts.Parents/caregivers also can view and update the registered children's information.

6.3 Autism Screening Management

The system provides a digital M-CHAT-R questionnaire consisting of 20 questions.Parents/caregivers can easily complete the questionnaire by answering Yes or No based on their children's behaviour.The system automatically calculates screening scores and categorizes the results into three risk levels according to the score they got which are Low Risk (0-2),Moderate Risk (3-7) and High Risk (8-20).Screening results are store in the database

6.4 Behaviour Monitoring Management

Parents/caregivers can record children's emotional conditions and behavioural activities.The system stores behavioural records including the dates and times.

6.5 Appointment and Referral Management

Parents/caregivers can view and manage appointment information with relevant specialists and clinics centres.The system stores appointment dates, times,and locations.The system provides appointments reminders and notifications for upcoming consultations and follow-up sessions in order to ensure it can be done smoothly.Appointment details will be kept in the database for easier tracking and retrieval.

6.6 Reporting and Analytics Management

The system generates reports based on stored children's profiles, screening results, and behavioural records.The system provides summarized information to support organizational monitoring and decision-making.

6.7 User Roles and Responsibilities

The proposed system consists of the following main user roles:

1. Parents / Caregivers

- Register and manage their accounts.
- Create, view, and update children's profiles.
- Complete M-CHAT-R screening questionnaires.
- View children's screening results.
- Record and monitor children's behavioural activities.
- Manage appointments and view referral information.
- Receive notifications and reminders.

2. NGO Administrators (System Owner)

- Manage and monitor registered user accounts.
- Access authorized children's records and screening information.
- Monitor overall screening activities.
- Generate reports and view analytical information.

3. Healthcare Specialists

- Manage appointment schedules
- Approve, reject, or reschedule appointment requests based on their availability.

6.8 System Limitations and Exclusions

The proposed system focuses on early screening support and information management. Therefore, the following activities are excluded from its scope:

- The system does not provide official autism diagnosis or replace professional medical assessments.
- The system does not provide medical treatment, therapy sessions, or prescribe medication.
- The system does not replace the official BPPOKU registration and approval process conducted by the relevant authorities.
- The system does not perform AI-based autism detection through video recording or facial recognition.
- The system does not manage financial transactions, medical payments, or insurance claims.

- The system does not guarantee the availability of appointments or services at external healthcare and educational facilities.

7. Project Planning

7.1 Member roles and responsibilities

Table 1: Member Roles and Responsibilities

Role	**Member name**	**Responsibilities**
Project Manager and Developer Leader	Balqis Batrisya binti Jalaluddin	-Leads planning phase and implementation phase -Manages the project schedule, GitHub repository, and team meetings -Develops the login and authentication setup and core website modules.
System analyst and QA Tester	Nasaaie bin Noriskamar	-Leads requirement analysis phase and testing phase -Prepares system diagrams (Structure Chart, Physical DFD) and database design -Creates test cases, checks for bugs, and writes testing reports.
UI/UX and System Designer	Yong See En	-Leads system design phase -Designs the website interface and wireframes using Figma -Builds the front-end pages and connects the system modules together.
Deployment and Documentation Leader	Han Tian Chou	-Leads deployment and increment phase -Launches and hosts the completed website online

		-Writes the user manual, quick user guide, and increment review report.
Integration & Security Leader	Qistina Batrisyia binti Noor Mohd Azlan	-Leads final integration and maintenance phase -Checks that all system features work smoothly together and data is secure -Compiles the final documentation and project sign-off report.

7.2 Planning for each phase

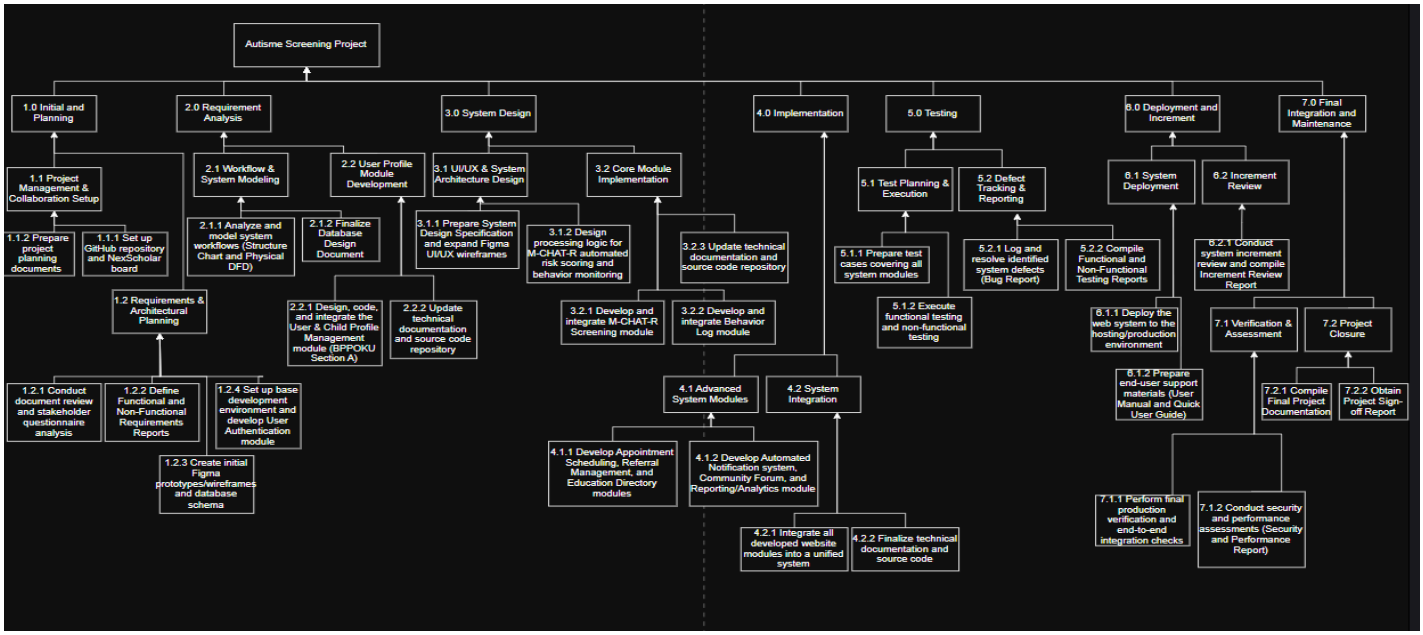
Table 2: Planning of Each Phase

Phase	**Duration**	**Person-In-Charge (as leader of the sprint)**	**Sprint Goal**	**Deliverables**
Phase 1: Initial and Planning	Week 1-2 (2week)	Balqis Batrisya binti Jalaluddin	Project setup, initial requirements, developing first module(user authentication and base environment)	-Team Setup Documentation (GitHub Repository, NexScholar Board) -Project Planning Documents (Feasibility Study Report, Project Proposal, Project Schedule) -Document Review Report -Questionnaire Report -Functional Requirements and Non-Functional

				Requirements Reports -Figma Prototype/Wireframes -GitHub Repository -Database Schema -Source Code -Developed Website Modules
Phase 2: Requirement Analysis	Week 3-4 (2 weeks)	Nasaaie bin Noriskamar	Designing, implementing and integrating the first main functional module(user module)	-Structure Chart -Physical DFD -Database Design Document -Source Code -Developed Website Modules -Integrated System Components -Technical Documentation
Phase 3: System Design	Week 5-6 (2 weeks)	Yong See En	Expanding system modules and processing logic	-System Design Specification -Figma Prototype/Wireframes -Source Code -Developed Website Modules -Integrated System Components -Technical Documentation
Phase 4: Implementation	Week 7-8 (2 weeks)	Balqis Batrisya binti Jalaluddin	Completing remaining website modules and report documentation	-Source Code -Developed Website Modules -Integrated System Components -Technical Documentation

Phase 5: Testing	Week 9-10 (2 week)	Nasaaie bin Noriskamar	Testing and reporting	-Test Cases -Bug Report -Functional Testing Report and Non-Functional Testing Report
Phase 6: Deployment and Increment	Week 11-12 (2 week)	Han Tian Chou	Launching system and preparing end-user supports	-Deployed System -User Supports including user manual and quick user guide implementing -Increment Review Report
Phase 7: Final Integration and Maintenance	Week 13-14 (2 week)	1)Qistina Batrisyia binti Noor Mohd Azlan	Final production verification, security assessment and project sign-off	-Final Integrated System -Security and Performance Report -Final Documentation -Project Sign-off Report

7.3 Work Breakdown Structure(WBS)



7.4 Project Planning Gantt Chart

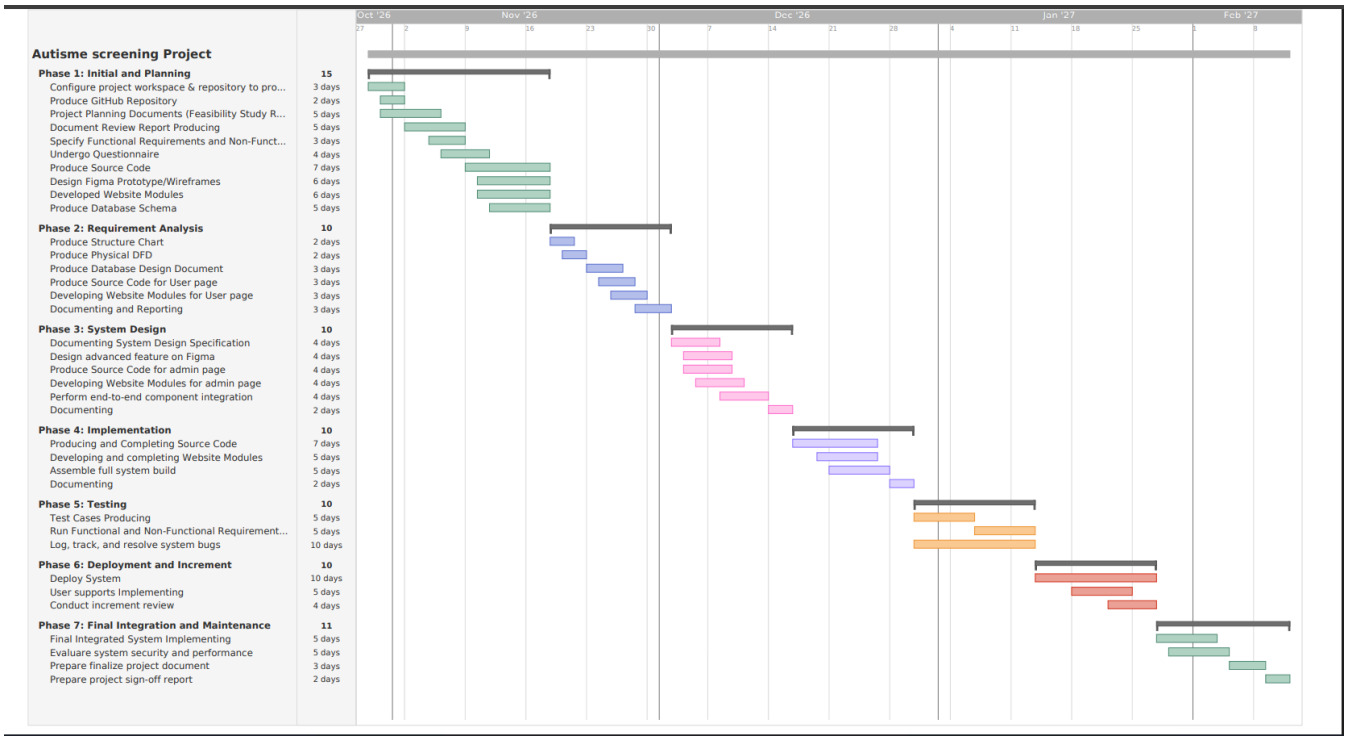


Figure 1: Project planning Gantt Chart

8. Requirement Analysis

8.1 *AS-IS* Process Flow

Nowadays, the current system relies heavily on manual processes and traditional methods. The current autism early screening process is primarily conducted through healthcare professionals, educators or autism specialists. Parents or guardians who suspect developmental delays in their children are required to visit a healthcare centre, clinic or autism intervention centre for an initial assessment.

During the screening session, parents are required to complete paper-based questionnaires which are known as M-CHAT-R regarding the child's communication skills, social interaction, repetitive behaviours, and developmental milestones. Healthcare professionals then observe the child's behaviour and review the completed questionnaire together with the parents.

Based on the collected information, the healthcare professional performs a preliminary screening to determine whether the child shows signs related to Autism Spectrum Disorder (ASD). If the screening shows possible autism traits, the child is referred for a comprehensive diagnostic assessment conducted by qualified specialists. If the child is formally diagnosed with Autism Spectrum Disorder (ASD) by a qualified specialist, the parents or guardians may register the child with the Department of Social Welfare (Jabatan Kebajikan Masyarakat, JKM). To complete the registration, they are required to submit the BPPOKU form together with the necessary supporting documents, such as the medical diagnosis report and identification documents. Upon successful registration, the child may become eligible for welfare assistance, support services, and benefits provided by JKM.

The current process has several limitations, where the screening process is time-consuming, relies heavily on manual documentation, and requires parents to attend physical appointments. In addition, there is limited accessibility to specialists in certain areas and parents often experience delays before receiving screening results or recommendations for further assessment.

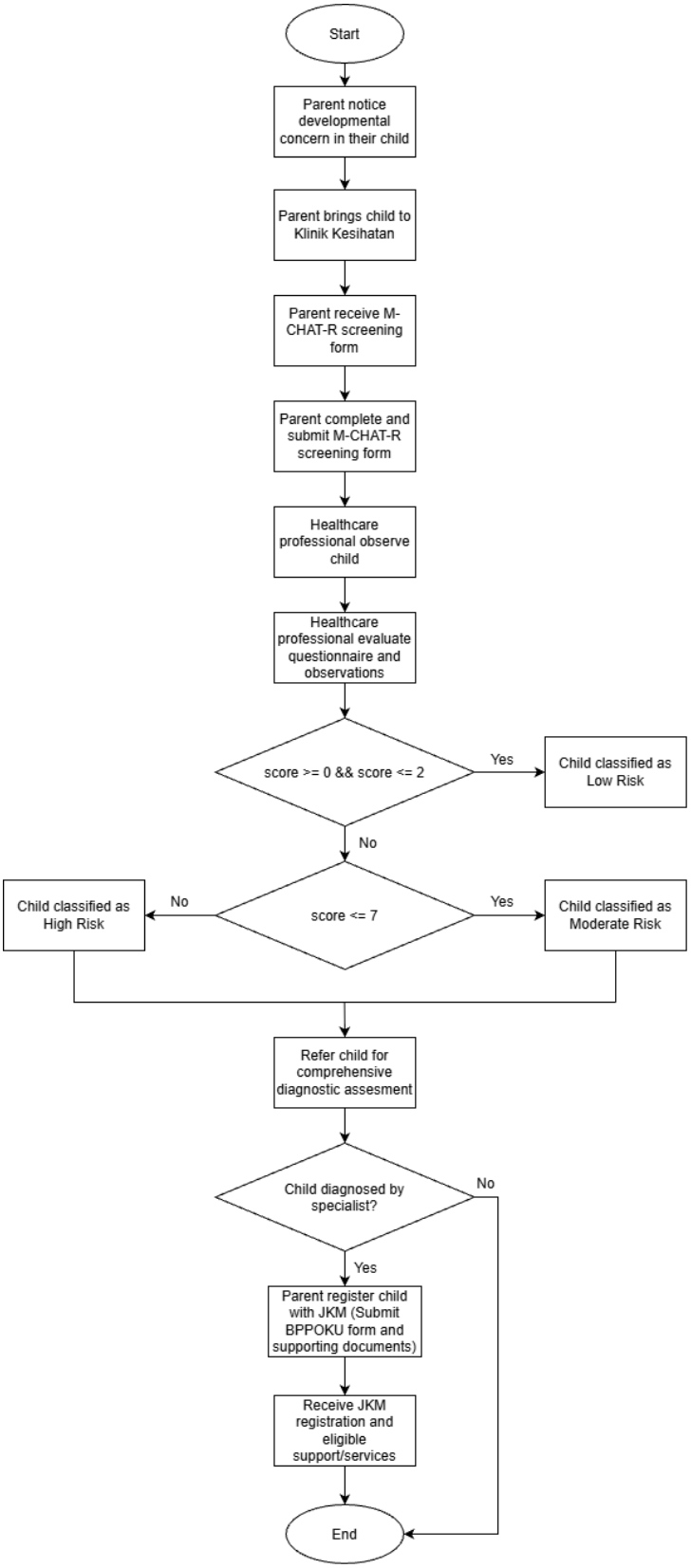


Figure 2: Workflow Flowchart

8.2 Business Rule

The existing autism early screening process follows several rules and guidelines to ensure screening results are evaluated properly. The screening questionnaire that is used in the process is M-CHAT-R and it consists of 20 questions. Each question only has 2 answer choices which is “Yes” or “No” and each answer has a different risk scoring method according to the question number. If the answer choice for number 2, 5 and 12 is “Yes” and all other questions are “No”, it will show a risk score. The total of the score will determine the child’s risk category where score 0-2 is categorized as Low Risk, score 3-7 is categorized as Moderate Risk and score 8-20 is categorized as High Risk.

Children in the Low Risk category do not require follow-up action unless the signs of Autism Spectrum Disorder (ASD) risk are observed, and children of age under 24 months, who are in the Low Risk category need to go through rescreening when they reach the age of 24 months. Next, children in the Moderate Risk category must complete the M-CHAT-R follow-up assessment. If the score is 2 or higher, the child is considered positive and should refer for diagnostic evaluation and eligibility assessment for early intervention, while children in the High Risk category can directly refer for diagnostic evaluation without having a follow-up assessment.

For Jabatan Kebajikan Masyarakat (JKM), a child must receive a formal ASD diagnosis from a qualified healthcare specialist before proceeding with registration, and BPPOKU form needs to be completed for the registration purposes. In addition, medical diagnosis reports and identification documents are also needed as supporting documents. Successful registration may qualify the child for welfare assistance, support services and other benefits that are provided by JKM.

8.3 Context Diagram

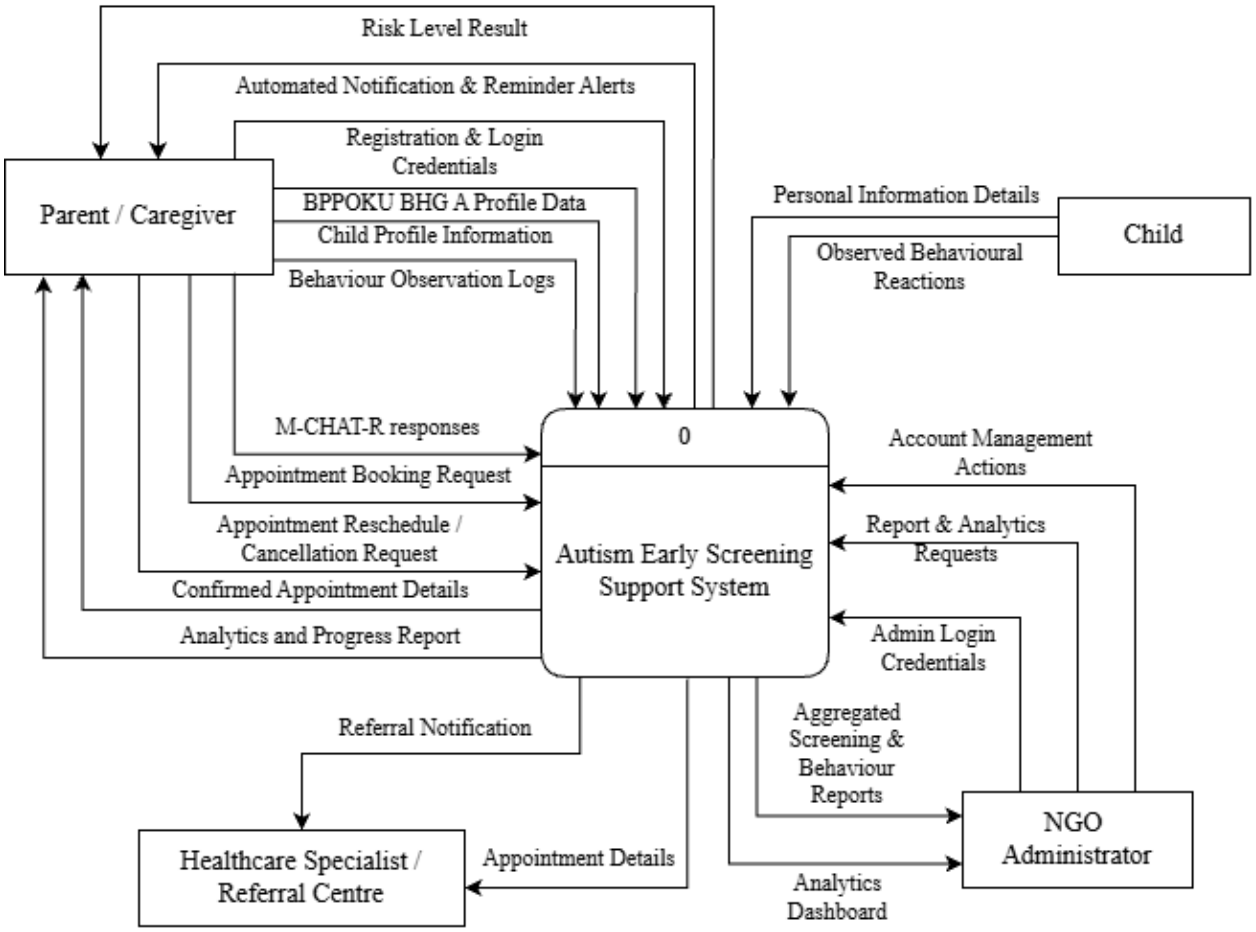


Figure 3: To-Be Context Diagram

8.4 Diagram 0

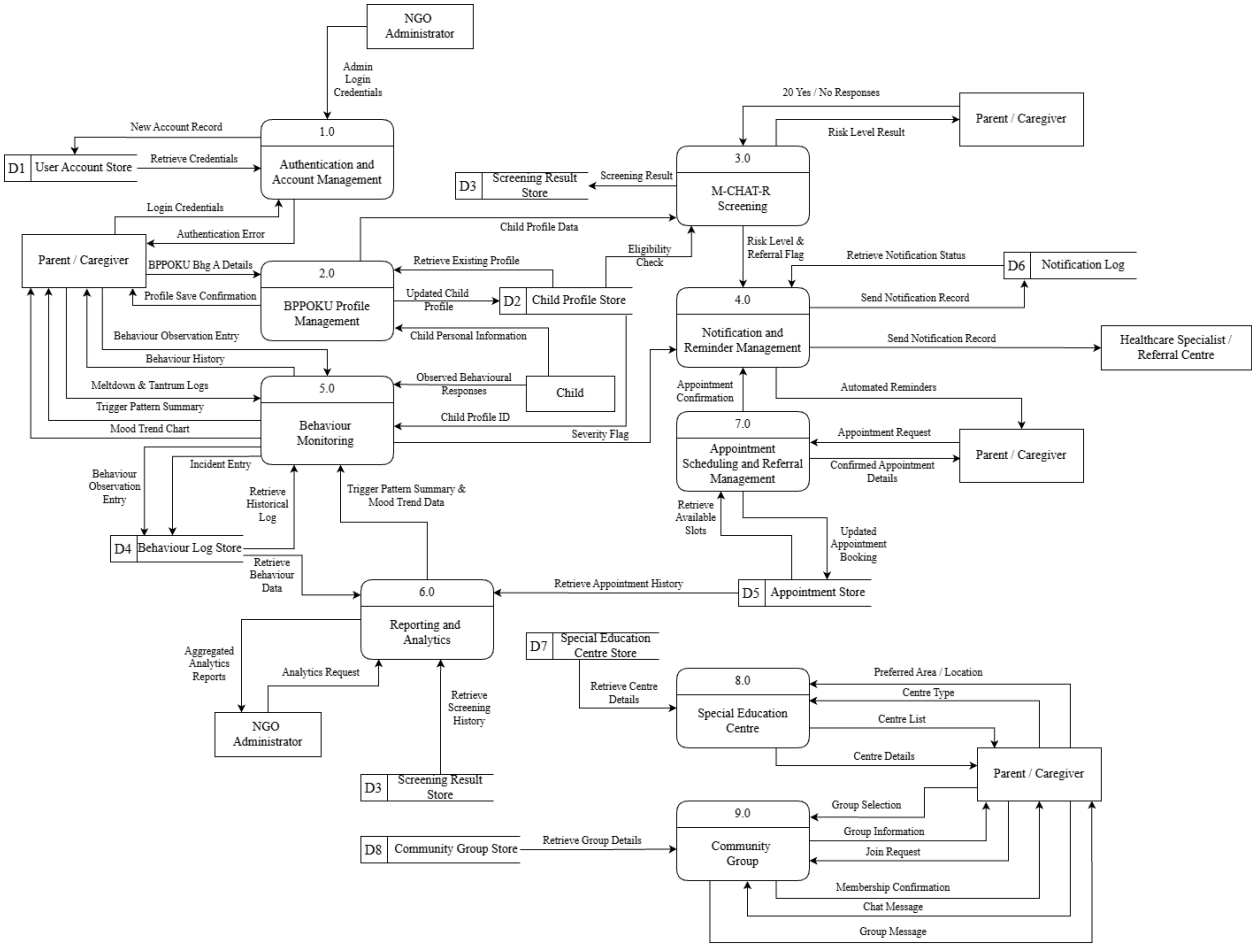


Figure 4: To-Be Diagram 0

9. Benefits of Proposed System

9.1 Improved Data Management & Centralization

Through the proposed system, the usage of physical filling M-CHAT-R questionnaires, handwritten BPPOKU forms and manual appointment diaries will be replaced by a centralized system that consists of all these functions and all database related autism. This transforms massive, scattered streams of data into structured digestible formats, eliminating clutter in the system. Hence, time consuming on finding and managing autism files can be reduced significantly. Adopting centralization can ease admin jobs on managing and transmission of the required data. Besides, storing files in a digital repository can ensure the records are never misplaced, duplicated, damaged or missing. Additionally, the new system can enhance data security and privacy as only authorized personnel can access their certain parts.Therefore, it can ensure no data leaked or misused.

9.2 Reduced Human and Scoring Errors

Through automated M-CHAT-R risk scoring, mistakes taken on calculating the score manually can be eliminated, ensuring a more reliable screening results. The new system consistently applies standardized thresholds to classify the severity of the autism condition. The transcription errors such as spelling mistakes can be avoided especially when the NGO staff need to manually transcribe the handwritten forms. In addition, parents can check on the screening, reporting, appointments at any time rather than relying on memorizing it. The system even provides a notification for scheduled events.

9.3 Faster Information Retrieval & Decision-Making

Through the new system, NGO staff and parents can retrieve any files within seconds instead of searching through physical folders. Not only that, staff or admin can evaluate and handle their task in a faster and more convenient way as the system provides built-in report and analytic tools. Therefore, human resources consumed on managing can be minimized. For the parents, the new systems can handle screening and give results instantly.Then, a quick search and filter on nearby specialized education centres can be taken, allowing parents to take immediate intervention.

9.4 Increased Operational Efficiency & Timely Intervention

Through the new system, the appointment and referral turnaround can be accelerated as phone calls and manual diary checks will be replaced by instant booking that provides real-time slot availability checks and automated digital referral letter generation.Other than that, the operational work to manually remember and message each parent will be replaced by the automated notification. The new system acts as a bridge to connect parents, NGO and specialists

using the community features. Any reminder and information can be shared through forum, reducing caregiver isolation.

10.Conclusion

In conclusion, the Autism Early Screening Support System is proposing a centralized database application to achieve a better version than the existing early autism screening. The proposed system addresses several problems regarding old school methods, including scattered records, manual paperwork, difficulties in retrieving data, and delays in screening follow-ups.
In addition, the proposed database application is expected to achieve a higher level of data organization, easier information retrieval,and support better monitoring of children's behavioural information. The stored information can also assist NGO administrators in monitoring screening activities and make a good decision on it
Overall, this system is expected to contribute to Excellent Synergy Resources by improving its information management capabilities and supporting more efficient early screening activities. It also provides better coordination between parents, caregivers, NGOs, and healthcare specialists.